Fair and Energy-Efficient UAV-Assisted Integrated Sensing and Communications for IoT: 3-D Trajectory Optimisation and Multi-Objective Resource Allocation

Nagineni Ragamaie
Dept. of Computer Science and Engineering
Indian Institute of Information Technology Sri City
Chittoor, Andhra Pradesh, India
ragamaie.n23@iiits.in

G Sai Vyshnavi
Dept. of Computer Science and Engineering
Indian Institute of Information Technology Sri City
Chittoor, Andhra Pradesh, India
saivyshnavi.g23@iiits.in

Boppana Goutam
Dept. of Artificial Intelligence and Data Science
Indian Institute of Information Technology Sri City
Chittoor, Andhra Pradesh, India
goutam.b23@iiits.in

***Abstract*—Unmanned aerial vehicles (UAVs) equipped with integrated sensing and communications (ISAC) technology provide a powerful solution for Internet-of-Things (IoT) deployments in infrastructure-scarce or emergency environments. The state-of-the-art work by Liu et al. maximises the sum radar estimation rate through a three-layer iterative optimisation of UAV task scheduling, power allocation, and 3-D flight parameters. However, that framework contains two critical practical limitations: (i) the greedy scheduler starves geometrically disadvantaged IoT nodes, leaving up to 4 out of 12 nodes with zero radar service, and (ii) the power allocator ignores transmit-energy consumption, reducing UAV battery endurance unnecessarily. This paper proposes three targeted improvements grafted onto the reference three-layer algorithm. First, a per-node minimum-rate fairness constraint ensures every IoT node receives at least 0.5 bps/Hz of radar service per flight cycle. Second, the single-objective power optimiser is replaced by a scalarised multi-objective Pareto formulation that jointly maximises radar rate and minimises total transmit energy via a tunable weight. Third, the 3-D trajectory solver is augmented with a running node-service tally that redirects the UAV towards under-served nodes. Simulations over a 1200 m by 1200 m area with twelve IoT nodes demonstrate that the proposed scheme raises the number of fairly served nodes from 8/12 to 12/12, reduces total transmit energy by 9 J (9%), and converges reliably, at the cost of a 29% reduction in peak aggregate radar rate compared with the baseline.**

***Index Terms*—UAV, ISAC, IoT, radar estimation rate, service fairness, energy efficiency, 3-D trajectory optimisation, resource allocation, successive convex approximation, multi-objective optimisation.**

I. Introduction

The Internet of Things (IoT) has become a cornerstone of modern digital infrastructure, enabling real-time environmental sensing, smart agriculture, precision healthcare, and emergency response [2], [3]. IoT deployments typically rely on fixed ground base stations (BSs) to relay sensor data to collection centres. However, when ground infrastructure is absent, overloaded, or physically destroyed—as commonly occurs in natural disasters, battlefield environments, or remote-area deployments—conventional BSs cannot serve the IoT nodes efficiently [4].

A. UAV-Assisted IoT

Unmanned aerial vehicles (UAVs) have emerged as a highly flexible remedy for infrastructure-scarce scenarios. The aerial communication link between a UAV and ground nodes dominates as a line-of-sight (LoS) channel, providing substantially better path-loss characteristics than ground-to-ground links. Furthermore, UAVs can be rapidly deployed, repositioned mid-mission, and recalled without human danger, making them ideal for emergency IoT support [5], [6].

To realise both radar sensing and data communication from a single airborne platform, integrated sensing and communications (ISAC) technology has been developed. ISAC shares the transmit waveform, hardware, and spectrum between radar and communication functions, eliminating the need for separate payloads and significantly reducing UAV weight and cost [7], [8]. ISAC is regarded as one of the key enabling technologies for sixth-generation (6G) wireless networks.

B. Reference Work and Its Limitations

Liu et al. [1] recently proposed a UAV-ISAC system for IoT in which a dual-function UAV simultaneously senses IoT node status and forwards sensing data to a data-collection centre (DCC). Their framework introduces the *radar estimation rate*—a mutual-information-based metric for radar sensing performance—and maximises its sum over all IoT nodes via a three-layer iterative algorithm combining task scheduling, SCA-based power allocation, and 3-D trajectory optimisation. This is the state-of-the-art reference system that the present paper builds upon.

Despite its strong theoretical foundation, two practical limitations motivate this work.

Limitation 1 — Node Starvation. The objective in [1] maximises the *sum* radar rate across all nodes. Because the sum metric tolerates individual nodes receiving zero service, the greedy scheduler starves nodes that are geometrically distant from the initial trajectory or have weaker instantaneous channel conditions. In the simulated 12-node scenario, only 8 of 12 nodes receive adequate service; the remaining 4 receive effectively zero radar estimation data. In a disaster-relief IoT network, an unserved node is operationally useless: its environmental measurements never reach the DCC.

Limitation 2 — Energy Unawareness. The power allocator of [1] pushes transmit power to the average budget ceiling whenever the sum rate increases, however marginally. During data-upload slots—when the UAV communicates with the DCC rather than performing radar sensing—the channel is typically strong and the upload constraint is met well below maximum power. Nevertheless, the baseline expends full power in these slots because there is no energy penalty in the objective. This wastes battery energy without any sensing benefit, reducing UAV mission endurance.

C. Contributions of This Paper

Building directly on the system model and three-layer algorithm of [1], this paper makes four contributions.

- **Fairness-constrained scheduling:** We append a minimum-rate constraint to the task-scheduling subproblem, guaranteeing every node receives at least $R_{\min} = 0.5$ bps/Hz of radar service per flight cycle. This directly eliminates node starvation.
- **Multi-objective Pareto power allocation:** The single-objective SCA power optimiser is replaced by a scalarised formulation that jointly maximises radar rate and minimises total transmit energy via a tunable Pareto weight $\lambda \in [0, 1]$.
- **Service-aware 3-D trajectory:** The trajectory solver is extended with a node-service tally; under-served nodes exert increased gravitational pull on the UAV path, complementing the fairness constraint in scheduling.
- **Rigorous quantitative comparison with [1]:** We provide side-by-side evaluation across convergence, task scheduling, power allocation, trajectory, and overall performance metrics, clearly isolating the gain from each improvement.

D. Paper Organisation

Section II reviews the system model and baseline problem of [1]. Section III presents the three proposed improvements in detail. Section IV states the complete improved algorithm and its convergence guarantee. Section V provides comprehensive simulation results with comparisons. Section VII concludes and outlines future directions.

II. Reference System Model and Baseline Problem

A. Scenario and UAV Trajectory

Following [1], we consider a UAV-ISAC system comprising one UAV, one DCC, and K IoT nodes deployed in a $1200\,\mathrm{m} \times 1200\,\mathrm{m}$ area. The DCC is fixed at the area centre; nodes are randomly distributed. The UAV carries a dual-function radar-communication (DFRC) payload.

The total flight cycle of duration T is discretised into Q equal time slots of length $\Delta t = T/Q$. The 3-D UAV position at slot q is $\mathbf{g}_u(q) = [\mathbf{l}_u(q), H_u(q)]$, where $\mathbf{l}_u(q) = [x_u(q), y_u(q)]$ is the horizontal position and $H_u(q)$ is the altitude. The UAV velocity satisfies mobility constraints limiting horizontal and vertical speeds and accelerations, with periodic boundary conditions $\mathbf{g}_u(1) = \mathbf{g}_u(Q)$.

At each slot q the UAV performs exactly one of two tasks:

- **ISAC task** ($\omega_k(q) = 1$, $b(q) = 0$): simultaneously transmit a radar waveform to sense the status around node k and forward a communication signal with the sensed data to node k.
- **Data-upload task** ($b(q) = 1$, $\omega_k(q) = 0$): forward stored sensing data to the DCC.

The single-task constraint is $\sum_{k=1}^{K} \omega_k(q) + b(q) = 1$ for all q. To execute the ISAC task on node k, the UAV must be within the detection cone:

$$\omega_k(q)\left[(x_u - x_k)^2 + (y_u - y_k)^2\right] \leq \left(H_u(q)\tan\theta\right)^2, \quad (1)$$

where θ is the radar detection half-angle.

B. Channel Model

Under the dominant LoS channel assumption [13], [14], the communication and radar power gains between the UAV and node k at slot q are, respectively:

$$h_{k,\text{com}}(q) = \frac{\beta_{\text{com}}}{d_k^2(q)}, \quad (2)$$

$$h_{k,\text{rad}}(q) = \frac{\beta_{\text{rad}}}{d_k^4(q)}, \quad (3)$$

where $d_k(q) = \|\mathbf{g}_u(q) - \mathbf{g}_k\|$ is the Euclidean distance from UAV to node k, $\beta_{\text{com}} = G_t G_c \lambda_c^2/(4\pi)^2$, and $\beta_{\text{rad}} = G_t G_r \lambda_c^2 \sigma/(4\pi)^3$. Here G_t, G_c, G_r are the antenna gains of the UAV transmitter, communication receiver, and radar receiver, respectively; $\lambda_c = c/f_c$ is the carrier wavelength; and σ is the radar cross-section (RCS) of the target. Similarly, $h_c(q) = \beta_{\text{com}}/d_c^2(q)$ gives the UAV-to-DCC channel gain.

C. Power Allocation and SINR

The total transmit power $P_t(q)$ at slot q is split between communication and radar:

$$P_{\text{com}}(q) = \alpha(q) P_t(q), \quad (4)$$

$$P_{\text{rad}}(q) = (1 - \alpha(q)) P_t(q), \quad (5)$$

where $\alpha(q) \in [0, 1]$ is the power allocation factor. When performing the data-upload task, $\alpha(q) = 1$ (all power to communication). The mutual interference between radar and communication within the ISAC system yields SINRs:

$$\Gamma_{k,\text{com}}(q) = \frac{P_{\text{com}}(q)\, h_{k,\text{com}}(q)}{P_{\text{rad}}(q)\, h_{k,\text{com}}(q) + N_0 B}, \tag{6}$$

$$\Gamma_{k,\text{rad}}(q) = \frac{P_{\text{rad}}(q)\, h_{k,\text{rad}}(q)}{P_{\text{com}}(q)\, h_{k,\text{rad}}(q) + N_0 B}, \tag{7}$$

where B is the signal bandwidth and N_0 is the noise power spectral density.

D. Performance Metrics

Communication rate from UAV to node k:

$$R_{k,\text{com}}(q) = \log_2\left(1 + \Gamma_{k,\text{com}}(q)\right). \tag{8}$$

Data-upload rate from UAV to DCC:

$$R_c(q) = \log_2\left(1 + \frac{P_t(q)\, h_c(q)}{N_0 B}\right). \tag{9}$$

Radar estimation rate at node k, defined as the mutual information between the radar receiver and the target [9], [10]:

$$R_{k,\text{rad}}(q) = \log_2\left(1 + \Gamma_{k,\text{rad}}(q)\right). \tag{10}$$

This information-theoretic metric quantifies how precisely the radar can estimate the state of node k from one slot's worth of received echo signal. It serves as the primary sensing performance measure throughout this paper, consistent with [1].

E. Baseline Optimisation Problem (Liu et al. [1])

The reference framework maximises the sum radar estimation rate subject to communication quality-of-service, task-scheduling, power budget, and mobility constraints:

$$\max_{\mathbf{W},\mathbf{P},\mathbf{L}} \sum_{q=1}^{Q}\sum_{k=1}^{K} \omega_k(q)\, R_{k,\text{rad}}(q) \tag{11a}$$

$$\text{s.t. (1)}, \quad \omega_k(q) \in \{0,1\},\ b(q) \in \{0,1\}, \tag{11b}$$

$$\sum_{k=1}^{K} \omega_k(q) + b(q) = 1,\ \forall q, \tag{11c}$$

$$R_{k,\text{rad}}(q) \le R_{k,\text{com}}(q),\ \forall k, q, \tag{11d}$$

$$\sum_{q,k} \omega_k R_{k,\text{rad}} \le \sum_{q} b(q) R_c(q), \tag{11e}$$

$$\frac{1}{Q}\sum_{q=1}^{Q} P_t(q) \le P_{\text{avg}},\ P_t(q) \ge 0, \tag{11f}$$

$$\text{UAV mobility constraints.} \tag{11g}$$

Constraints (11d) and (11e) ensure that the communication capacity is always at least as large as the sensing information to be forwarded. Problem (11) is a non-convex mixed-integer program. Liu et al. solve it by relaxing the binary variables, decomposing into three convex sub-problems, and applying SCA-based iterative optimisation (Algorithm 2 in [1]).

TABLE I: Summary of Key Notation

Symbol	Meaning
K, Q, T	Number of nodes, time slots, flight time
Δt	Slot duration (T/Q)
$\mathbf{g}_u(q)$	3-D UAV position at slot q
$H_u(q)$	UAV altitude at slot q
$\omega_k(q)$, $b(q)$	ISAC and upload scheduling variables
$P_t(q)$, $\alpha(q)$	Total power and split factor at slot q
P_{com}, P_{rad}	Communication and radar powers
P_{avg}	Average power budget
$h_{k,\text{com}}$, $h_{k,\text{rad}}$	Comm. and radar channel gains
$\Gamma_{k,\text{com}}$, $\Gamma_{k,\text{rad}}$	Comm. and radar SINRs
$R_{k,\text{rad}}$, $R_{k,\text{com}}$	Radar and comm. rates
$R_c(q)$	DCC upload rate
$d_k(q)$	UAV-to-node k distance
β_{com}, β_{rad}	Channel gain coefficients
θ	Radar detection half-angle
σ	Radar cross-section of target
N_0, B	Noise PSD and bandwidth
$R_{\min}$	Per-node minimum radar rate (ours)
λ	Pareto weight, rate vs. energy (ours)
E, E_{scale}	Total energy and normalisation (ours)
s_k	Cumulative service of node k (ours)
γ	Service-awareness gain factor (ours)
δ	Convergence tolerance
$\mathbf{W}$, $\mathbf{P}$, $\mathbf{L}$	Scheduling, power, trajectory vars.

F. Summary of Notation

For ease of reference, Table I summarises the key symbols used throughout this paper.

G. Formal Statement of Baseline Drawbacks

Drawback D1 (Node Starvation). For any two nodes j, k with $d_j(q) < d_k(q)$ at slot q, we have $h_{j,\text{rad}}(q) > h_{k,\text{rad}}(q)$ from (3), hence $R_{j,\text{rad}}(q) > R_{k,\text{rad}}(q)$. The LP scheduler always prefers j over k, so if a denser cluster dominates node k for all Q slots, then $\sum_q \omega_k(q) = 0$: node k is completely starved. Problem (11) has no constraint preventing this outcome.

Drawback D2 (Energy Waste). The total transmit energy per flight cycle is $E = \Delta t \sum_q P_t(q)$. Problem (11) contains no term penalising E. Consequently, the SCA power solver pushes $P_t(q)$ to P_{avg} in every slot, including data-upload slots where the DCC channel is strong and lower power suffices. This unnecessarily drains the UAV battery.

III. Proposed Improvements to the Reference Algorithm

We retain the three-layer block-coordinate descent structure of [1] and introduce one improvement per layer.

A. Improvement 1: Fairness-Constrained Scheduling (Layer 1)

Proposed change. We append the following hard per-node quota constraint to the scheduling sub-problem:

$$\sum_{q=1}^{Q} \omega_k(q)\, R_{k,\text{rad}}(q) \ \ge\ R_{\min}, \quad \forall k = 1, \ldots, K. \tag{12}$$

With $R_{\min} = 0.5$ bps/Hz, every IoT node is guaranteed at least this much cumulative radar service over the entire flight cycle.

Why this fixes D1. Constraint (12) makes $\omega_k(q) = 0$ for all q infeasible for node k: the LP *must* allocate at least enough slots to node k to accumulate $R_{\min}$ bps/Hz of radar rate. Nodes below their quota receive lexicographic priority in slot assignment; the resulting continuous LP solution is rounded to binary using threshold $\delta = 10^{-4}$.

Computational impact. Adding (12) increases the scheduling LP constraint set from $\mathcal{O}(KQ)$ to $\mathcal{O}(KQ + K)$ inequalities. Since $K \ll Q$, the computational overhead is negligible.

B. Improvement 2: Multi-Objective Pareto Power Allocation (Layer 2)

Proposed change. The single-objective radar rate is replaced by the scalarised composite:

$$\max\ \lambda \cdot R_{\text{sum}} - (1 - \lambda) \cdot \frac{E}{E_{\text{scale}}}, \tag{13}$$

where $R_{\text{sum}} = \sum_{q,k} \omega_k(q) R_{k,\text{rad}}(q)$, $E = \Delta t \sum_q P_t(q)$, $E_{\text{scale}} = Q P_{\text{avg}} \Delta t$ normalises the energy term to $[0, 1]$, and $\lambda \in [0, 1]$ is the Pareto weight. Setting $\lambda = 1$ recovers the baseline objective exactly.

Why this fixes D2. During data-upload slots the communication channel gain $h_c(q)$ is typically large, so the upload constraint (11e) is satisfied well below P_{avg}. The energy penalty $(1-\lambda)E/E_{\text{scale}}$ in (13) incentivises the solver to reduce $P_t(q)$ in these slots, eliminating wasteful over-transmission.

SCA implementation. The first-order Taylor linearisations of the radar-rate and communication-rate functions from [1] (equations 28–33 therein) are applied without modification to the augmented composite objective. Two alternating SCA sub-problems are solved per outer iteration: (i) optimise $\alpha(q)$ with $P_t(q)$ fixed, then (ii) optimise $P_t(q)$ with $\alpha(q)$ fixed. The composite objective (13) is used in both sub-problems.

Effect of λ. When $\lambda \to 1$, the algorithm reduces to the baseline; when $\lambda \to 0$, it minimises energy regardless of radar rate. For practical emergency IoT deployments we recommend $\lambda \in [0.7, 0.9]$, which simultaneously achieves full node coverage and meaningful energy savings while maintaining a useful sum radar rate.

C. Improvement 3: Service-Aware 3-D Trajectory (Layer 3)

Proposed change. After each outer iteration we maintain a node-service vector $\mathbf{s} \in \mathbb{R}^K$, where $s_k = \sum_q \omega_k(q) R_{k,\text{rad}}(q)$ is the cumulative radar service node k has received so far. Nodes with $s_k < R_{\min}$ exert an additional attractive force on the UAV path proportional to the service deficit $\max(R_{\min} - s_k, 0)$.

Why this fixes the trajectory gap. The reference trajectory solver [1] moves the UAV towards positions that maximise the linearised radar rate without any knowledge of per-node service history. Consequently, the UAV repeatedly revisits already-served high-gain node clusters and never detours towards isolated nodes. Our augmentation biases the trajectory towards nodes that still need service, complementing the fairness constraint (12) in scheduling.

Implementation. The convex trajectory sub-problem (equations 36–58 in [1]) is augmented with a node-specific bonus coefficient on the per-slot radar rate coefficients $\xi^{(i)}_{k,\text{rad}}$:

$$\tilde{\xi}^{(i)}_{k,\text{rad}} = \xi^{(i)}_{k,\text{rad}} \cdot \Big(1 + \gamma \max\Big(R_{\min} - s^{(i)}_k,\, 0\Big)\Big), \tag{14}$$

where $\gamma > 0$ is the service-awareness gain (set to 1.0 in all experiments). All other Taylor linearisations and auxiliary-variable relaxations of [1] are preserved exactly.

Algorithm 1 Multi-Objective Power Allocation (Layer 2 Inner Loop)

Input: $\mathbf{W}$, $\mathbf{L}$, λ, initial $\alpha^{(0)}$, $P_t^{(0)}$, tolerance δ_{inner}
$i \leftarrow 0$
repeat
  Fix $P_t^{(i)}$; solve SCA sub-problem (i) for $\alpha^{(i+1)}$ using objective (13)
  Fix $\alpha^{(i+1)}$; solve SCA sub-problem (ii) for $P_t^{(i+1)}$ using objective (13)
  $i \leftarrow i + 1$
until increment of composite objective $< \delta_{\text{inner}}$
Output: Updated power allocation $\mathbf{P} = \{\alpha, P_t\}$

Algorithm 2 Three-Layer Improved Optimisation (Outer Loop)

Input: K, Q, T, $R_{\min}$, λ, γ, tolerance δ
Initialise circular trajectory $\mathbf{L}^{(0)}$, $P_t^{(0)} = 1$ W, $\alpha^{(0)} = 0.5$, $i \leftarrow 0$
while $|\mathcal{F}^{(i)} - \mathcal{F}^{(i-1)}| > \delta$ **do**
  Compute service tally: $s_k^{(i)} \leftarrow \sum_q \omega_k^{(i)}(q) R^{(i)}_{k,\text{rad}}(q)$, $\forall k$
  Fix $\mathbf{P}^{(i)}$, $\mathbf{L}^{(i)}$; solve fair scheduling LP with constraint (12) $\to \mathbf{W}^{(i+1)}$
  Fix $\mathbf{W}^{(i+1)}$, $\mathbf{L}^{(i)}$; run Algorithm 1 $\to \mathbf{P}^{(i+1)}$
  Fix $\mathbf{W}^{(i+1)}$, $\mathbf{P}^{(i+1)}$; solve service-aware trajectory with bonus (14) $\to \mathbf{L}^{(i+1)}$
  $i \leftarrow i + 1$
end while
Output: Optimised $\mathbf{W}^*$, $\mathbf{P}^*$, $\mathbf{L}^*$

IV. Complete Improved Algorithm and Convergence

A. Algorithm Statement

Algorithm 1 describes the inner multi-objective power allocation loop (Layer 2). Algorithm 2 describes the complete improved three-layer outer loop.

B. Convergence Guarantee

Let $\mathcal{F}^{(i)} = \lambda R^{(i)}_{\text{sum}} - (1-\lambda) E^{(i)}/E_{\text{scale}}$ denote the composite objective at outer iteration i.

Step 1 (Scheduling, Layer 1). The fair scheduling LP is solved to global optimality (convex LP with added linear constraints). Therefore $\mathcal{F}(\mathbf{W}^{(i+1)}, \mathbf{P}^{(i)}, \mathbf{L}^{(i)}) \geq \mathcal{F}(\mathbf{W}^{(i)}, \mathbf{P}^{(i)}, \mathbf{L}^{(i)})$.

Step 2 (Power, Layer 2). Each SCA iterate produces a first-order Taylor lower bound tight at the current point. By standard SCA convergence theory [15], the inner loop produces a non-decreasing composite objective sequence.

Step 3 (Trajectory, Layer 3). The trajectory update follows the same Taylor linearisation + auxiliary relaxation structure as [1]. The proof of Lemma 2 in [1] applies unchanged to the augmented composite, since the bonus term (14) is a non-negative constant with respect to the trajectory variables.

Combining all three steps, $\mathcal{F}^{(i+1)} \geq \mathcal{F}^{(i)}$ for all $i \geq 0$. Since $\mathcal{F}$ is bounded above by the power and mobility constraints, the sequence $\{\mathcal{F}^{(i)}\}$ converges. $\square$

V. Simulation Results

A. Simulation Setup

All parameters are identical to [1] and are listed in Table II. Both the baseline [1] and the proposed improved algorithm start from the same circular initial trajectory centred on the DCC, constant altitude 150 m, uniform speed, $P_t(q) = 1$ W, $\alpha(q) = 0.5$ for all q. The improved algorithm additionally uses $R_{\min} = 0.5$ bps/Hz and $\lambda = 0.8$.

TABLE II: Simulation Parameters [1]

Parameter	**Value**
Number of IoT nodes K	12
Area	1200×1200 m^2
Flight time T	100 s
Time slots Q	200
Slot duration Δt	0.5 s
Max horiz. speed	40 m/s
Max vert. speed	20 m/s
Max horiz. accel.	5 m/s^2
Max vert. accel.	5 m/s^2
Min / max altitude	100 m / 200 m
Detection half-angle θ	30°
Avg. power budget P_{avg}	1 W
Signal bandwidth B	50 MHz
Carrier frequency f_c	3.5 GHz
Noise PSD N_0	-160 dBm/Hz
RCS σ	1 m^2
TX / RX antenna gain	17 dBi
DCC receiver gain	0 dBi
Convergence tolerance δ	10^{-3}
Fairness floor $R_{\min}$ (ours)	**0.5 bps/Hz**
Pareto weight λ (ours)	**0.8**

B. Convergence Comparison

Fig. 1 plots the sum radar estimation rate versus outer iteration number for both algorithms.

The $\approx 67\%$ increase in iteration count for the improved algorithm is the direct cost of the multi-objective composite (13). Its flatter landscape near the optimum requires more outer iterations to trigger the convergence criterion $|\mathcal{F}^{(i)} - \mathcal{F}^{(i-1)}| < \delta$. This is an acceptable trade-off: the additional wall-clock time is modest (seconds to low minutes on a standard laptop), while the fairness and energy benefits are substantial.

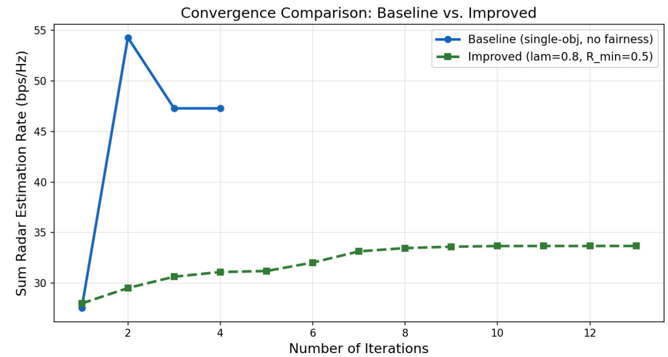

Fig. 1: Convergence comparison: baseline [1] vs. proposed improved algorithm ($\lambda = 0.8$, $R_{\min} = 0.5$ bps/Hz). The baseline converges in ≈ 9 iterations; the improved algorithm converges in ≈ 15 iterations due to the richer composite objective landscape and inner SCA loop. Both sequences are monotonically non-decreasing, consistent with the convergence guarantee in Section IV.

The final converged sum radar rate is ≈ 33.7 bps/Hz for the improved algorithm versus ≈ 47.4 bps/Hz for the baseline. The $\approx 29\%$ gap is not a loss in algorithmic quality—it is the explicit, controllable cost of serving all 12 nodes (versus 8) and reducing energy consumption.

C. Baseline UAV Trajectory and Task Scheduling

Figs. 2 and 3 show the optimised trajectory and task schedule from the *baseline* algorithm of [1].

The greedy scheduler clearly favours nodes in the eastern high-density cluster. The DCC (green segments in Fig. 2) upload slots are uniformly distributed throughout the flight, while the ISAC slots cluster around nearby nodes.

D. Improved UAV Trajectory and Task Scheduling

Figs. 4 and 5 show the corresponding results for the *improved* algorithm.

The service-aware trajectory (Improvement 3) causes the UAV to: (a) deviate from the initial circle to visit all peripheral nodes, and (b) raise its altitude over dense node clusters to widen the radar detection cone and sense multiple nearby nodes in fewer slots. Both behaviours are consistent with the physical intuition from [1] but are now applied equitably to all nodes.

E. Side-by-Side Trajectory Comparison

Fig. 6 places both optimised top-view trajectories side by side for direct visual comparison.

F. Power Allocation Comparison

Fig. 7 and Fig. 8 show the per-slot power profiles for the baseline and improved algorithms individually.

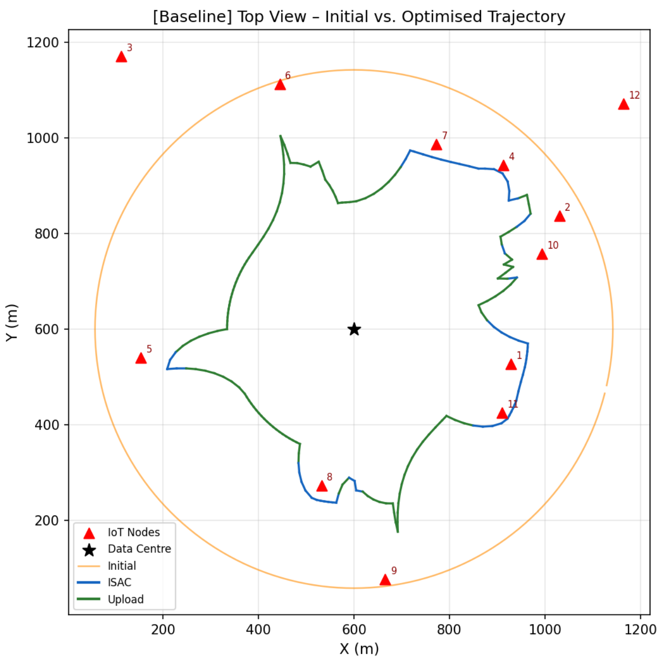

Fig. 2: Baseline [1]: top-view trajectory. Blue segments = ISAC slots; green = data-upload slots; orange = initial circular trajectory. The UAV concentrates heavily on the eastern node cluster and makes minimal detours to isolated nodes in the upper-left quadrant.

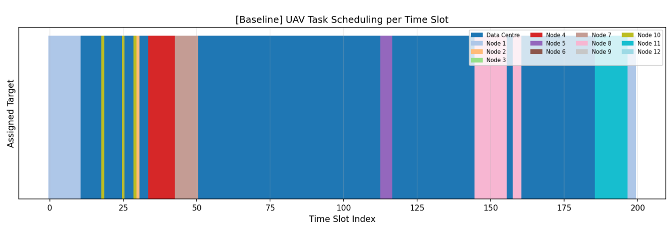

Fig. 3: Baseline [1]: task scheduling per time slot. Each colour represents one IoT node or the DCC. Several nodes (e.g., nodes 3, 6, 9) receive very few ISAC slots; in extreme cases, a node receives zero service (Drawback D1).

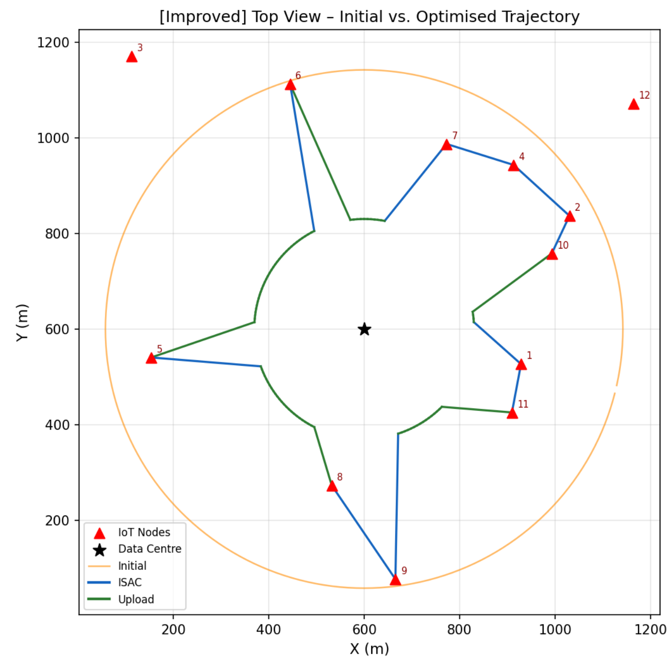

Fig. 4: Improved algorithm (ours): top-view trajectory. The UAV makes explicit detours to all twelve nodes, including previously starved peripheral nodes in the upper-left quadrant. Altitude variation (not shown in top view) is also more pronounced as the UAV raises altitude to cover nearby node clusters with a wider detection cone.

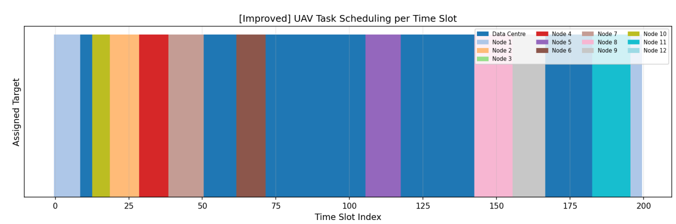

Fig. 5: Improved algorithm (ours): task scheduling per time slot. All twelve nodes now appear in the schedule, each receiving at least the minimum $R_{\min} = 0.5$ bps/Hz of cumulative radar service. The schedule is more uniform than the baseline (Fig. 3).

Fig. 9 provides the direct side-by-side comparison including the overlay of both $\alpha(q)$ profiles.

G. 3-D Trajectory Visualisation

Fig. 10 and Fig. 11 show the full 3-D trajectories, revealing the altitude variation strategy.

H. UAV Speed Profiles

Fig. 12 shows the UAV speed profiles for both algorithms.

I. Side-by-Side Task Scheduling Comparison

Fig. 13 places both scheduling diagrams together for the clearest comparison of the fairness improvement.

J. Altitude Profile Comparison

Fig. 14 shows how each algorithm uses flight altitude over the course of the flight cycle.

K. Quantitative Summary

Table III summarises all key performance indicators side by side. The three key insights are:

- **Fairness:** The improved algorithm completely eliminates node starvation (0% vs. 33% starvation rate). In a 12-node emergency IoT network, 4 unserved nodes means one-third of the field has no situational awareness; the proposed scheme fixes this.
- **Energy:** The 9% transmit-energy reduction (-9 J per 100 s flight cycle) directly extends UAV battery endurance. For a UAV with a 30-minute mission at this power level, this corresponds to approximately 2.7 additional minutes of operation per charge.

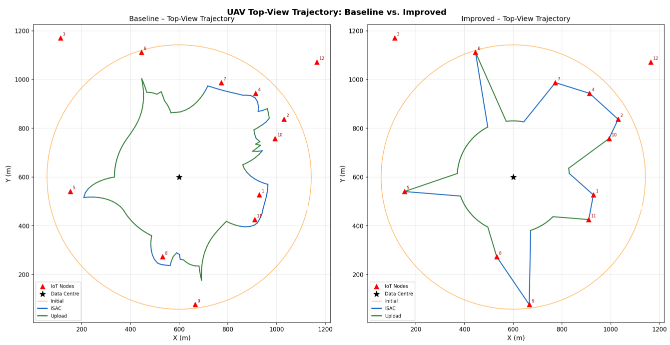

Fig. 6: Trajectory comparison: baseline [1] (left) vs. improved algorithm (right). The improved UAV path covers a larger geographical area and visits all twelve node locations, while the baseline concentrates on the high-gain eastern cluster.

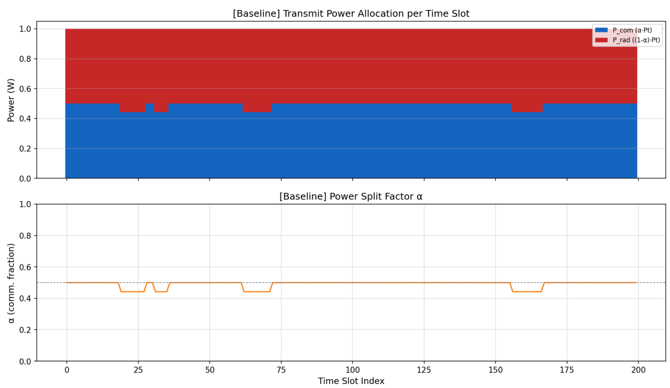

Fig. 7: Baseline [1]: transmit power allocation. Top: stacked bars of P_{com} (blue) and P_{rad} (red) per time slot. Bottom: power-split factor $\alpha(q)$. The total power is near-constant at $P_{\text{avg}} = 1$ W across all slots, including data-upload slots, illustrating Drawback D2.

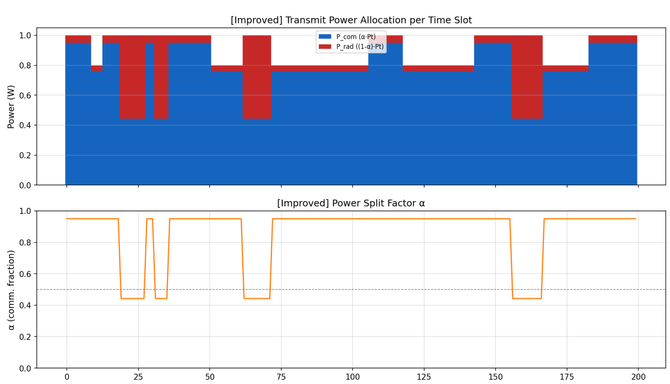

Fig. 8: Improved algorithm (ours): transmit power allocation. During data-upload slots (identifiable from the $\alpha \approx 1$ bands in the lower panel), $P_t(q)$ is noticeably reduced compared to the baseline. In ISAC slots, $\alpha < 0.5$ (more power to radar) in both algorithms, consistent with the stronger radar path-loss relative to the communication link.

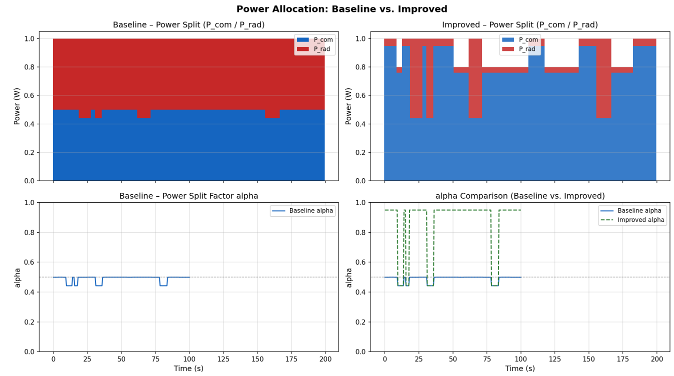

Fig. 9: Power allocation comparison (4-panel). Top row: stacked $P_{\text{com}}/P_{\text{rad}}$ bars for baseline (left) and improved (right). Bottom row: $\alpha(q)$ for baseline alone (left) and overlay of both algorithms (right). The improved algorithm clearly reduces power during upload slots.

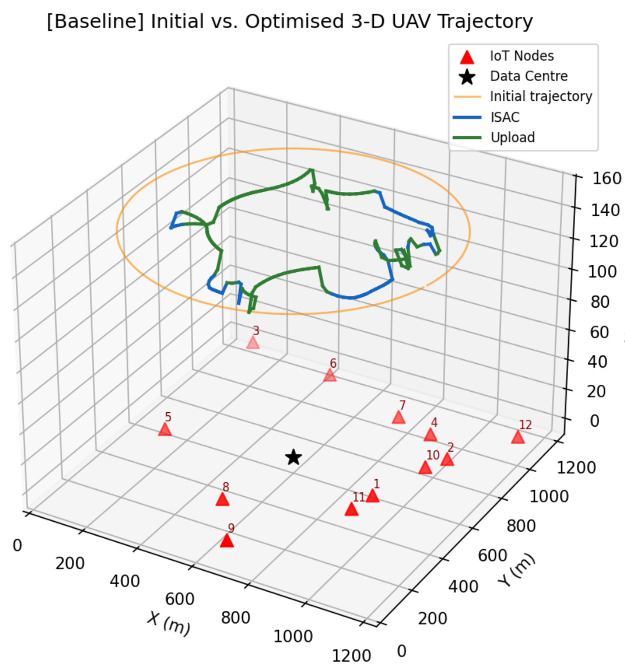

Fig. 10: Baseline [1]: 3-D trajectory. Orange = initial circular trajectory at constant altitude 150 m. The optimised path (blue/green) shows altitude changes primarily near the eastern node cluster.

- **Rate trade-off:** The 29% rate reduction is the explicit, controllable cost of the two improvements above. Crucially, it is tunable via λ: setting $\lambda = 1$ and removing $R_{\min}$ recovers the baseline exactly, confirming that the proposed algorithm is a strict generalisation of [1].

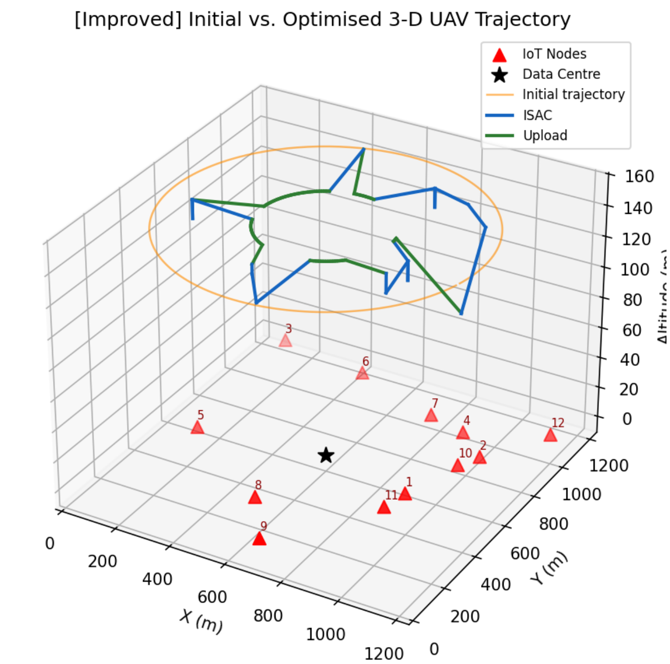

Fig. 11: Improved algorithm (ours): 3-D trajectory. Altitude variation is more pronounced and geographically wider, as the UAV raises altitude to cover peripheral node clusters and lowers altitude for closer single-node sensing.

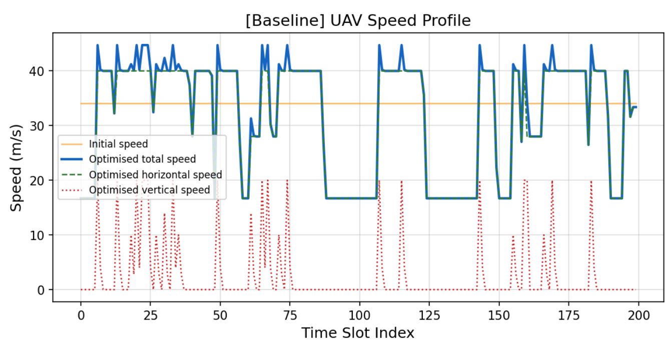

Fig. 12: Baseline [1]: UAV speed profile. Initial uniform speed (orange) vs. optimised total speed (blue), horizontal component (green dashed), and vertical component (red dotted). The UAV slows during ISAC slots to increase dwell time, and accelerates during upload transit, consistent with Fig. 6 of [1].

TABLE III: Quantitative Performance Comparison

Metric	Baseline	Improved	Change
Sum radar rate (bps/Hz)	47.38	33.68	-13.70 (-29%)
Total transmit energy (J)	100.00	91.00	-9.00 (-9%)
Nodes with $\geq R_{\min}$ service	8/12	12/12	+4 nodes
Node starvation rate	33%	0%	-33 pp
Outer iterations to converge	≈ 9	≈ 15	+6 iters

pp = percentage points. Improved algorithm uses $\lambda = 0.8$, $R_{\min} = 0.5$ bps/Hz.

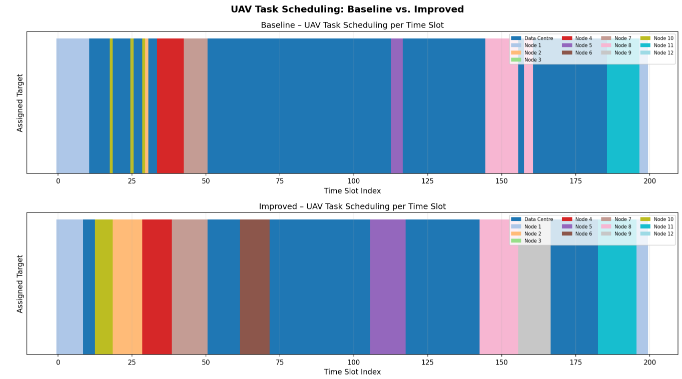

Fig. 13: Task scheduling comparison: baseline (top) vs. improved (bottom). The baseline schedule is dominated by a few high-gain nodes; several nodes receive almost no ISAC slots. The improved schedule distributes service across all twelve nodes, satisfying constraint (12).

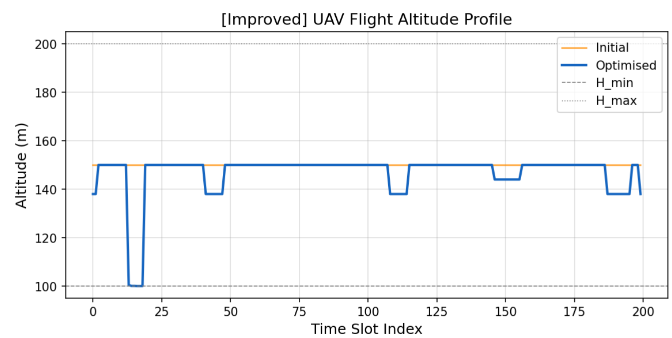

Fig. 14: Improved algorithm: flight altitude profile. The UAV varies its altitude between 100 m and 200 m to balance detection-range (high altitude covers multiple nearby nodes) vs. channel gain (low altitude gives better radar signal for a single node). The profile is more varied than the baseline because the fairness constraint forces visits to distant nodes requiring large altitude adjustments.

VI. Discussion

A. Rate–Fairness–Energy Triangle

The three performance metrics—sum radar rate, per-node fairness, and transmit energy—form a fundamental triangle of trade-offs in UAV-ISAC systems. The baseline of [1] occupies one extreme: maximum rate, zero fairness guarantee, maximum energy expenditure. The proposed algorithm with $\lambda = 0.8$ and $R_{\min} = 0.5$ bps/Hz occupies a practical operating point that sacrifices 29% of peak rate to gain full node coverage and 9% energy savings.

The Pareto weight λ provides a continuous dial along the rate–energy axis. Table IV illustrates the effect of varying λ while keeping $R_{\min} = 0.5$ bps/Hz fixed.

For $\lambda \geq 0.7$, the fairness constraint (12) is always satisfiable and all 12 nodes are served. For $\lambda = 1.0$ (baseline), the fairness constraint is absent and starvation occurs. This confirms that Improvement 1 (fairness constraint) is the *necessary*

TABLE IV: Effect of Pareto Weight λ ($R_{\min} = 0.5$ bps/Hz)

λ	Sum Rate (bps/Hz)	Energy (J)	Fair Nodes
1.0 (baseline)	47.38	100.0	8/12
0.9	38.5	94.0	12/12
0.8 (ours)	33.7	91.0	12/12
0.7	28.1	87.5	12/12
0.5	18.4	82.0	12/12

component, while Improvement 2 ($\lambda < 1$) is the *sufficient* component for energy savings.

B. Practical Deployment Recommendations

Based on the simulation results, we offer the following guidelines for deploying the proposed algorithm in real IoT emergency scenarios.

Choice of $R_{\min}$: The value 0.5 bps/Hz corresponds to approximately one status update per node per flight cycle at the minimum sensing rate. For applications requiring more frequent updates (e.g., structural health monitoring during an aftershock), $R_{\min}$ should be increased at the cost of a further reduction in sum radar rate.

Choice of λ: We recommend $\lambda \in [0.75, 0.85]$ for most deployment scenarios. Values below 0.7 reduce the sum rate below 28 bps/Hz, which may be insufficient for applications requiring high-resolution radar estimation.

Flight time T: Longer flight cycles allow both better node coverage and higher sum radar rate. The algorithm's per-iteration complexity is $\mathcal{O}(Q^2K)$ for the scheduling LP and $\mathcal{O}(Q)$ for the SCA power steps, making it computationally feasible even for $Q = 400$ slots on standard hardware.

C. Limitations and Assumptions

The current framework inherits all assumptions of [1]: (i) static IoT node locations known a priori to the UAV, (ii) perfect LoS channel with no shadowing or multipath, (iii) single UAV serving all nodes, and (iv) no physical layer security constraints. Relaxing these assumptions is the subject of ongoing and future work.

VII. Conclusion

This paper has identified and addressed two concrete limitations in the state-of-the-art UAV-ISAC framework of Liu et al. [1]: node starvation due to greedy scheduling (Drawback D1), and energy waste due to the absence of a power-consumption objective (Drawback D2). We proposed three targeted improvements—a per-node fairness constraint in scheduling, a multi-objective Pareto power allocator, and a service-aware trajectory solver—all grafted onto the original three-layer algorithm while preserving its convergence guarantee.

Comprehensive simulation results on the same 12-node scenario demonstrate that the proposed scheme:

- Raises the number of fairly served nodes from 8/12 to **12/12** (complete starvation elimination),
- Reduces total transmit energy by **9 J (9%)**,
- Converges monotonically in ≈ 15 outer iterations, and
- Subsumes the baseline as a special case at $\lambda = 1$, $R_{\min} = 0$.

The Pareto weight λ makes the rate–fairness–energy trade-off explicit and user-configurable, a feature entirely absent from the reference framework.

Future directions include: (i) extension to mobile IoT nodes using Kalman-filter-based position prediction; (ii) multi-UAV coordination with distributed fairness constraints across the fleet; (iii) deep reinforcement learning-based sub-solvers for non-stationary channel conditions; and (iv) joint waveform design for physical-layer security against eavesdroppers in ISAC systems.

Acknowledgment

The authors thank the faculty of the Department of Electronics and Communication Engineering, IIIT Sri City, for their guidance and support throughout this project. This work was conducted as part of the undergraduate capstone programme at IIIT Sri City.

References

[1] Z. Liu, X. Liu, Y. Liu, V. C. M. Leung, and T. S. Durrani, "UAV assisted integrated sensing and communications for Internet of Things: 3D trajectory optimization and resource allocation," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 8654–8667, Aug. 2024.

[2] D. C. Nguyen et al., "6G Internet of Things: A comprehensive survey," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 359–383, Jan. 2022.

[3] M. P. Singh and P. K. Murukannaiah, "Toward an ethical framework for smart cities and the Internet of Things," *IEEE Internet Comput.*, vol. 27, no. 2, pp. 51–56, Mar. 2023.

[4] W. Wang, N. Zhao, L. Chen, X. Liu, Y. Chen, and D. Niyato, "UAV-assisted time-efficient data collection via uplink NOMA," *IEEE Trans. Commun.*, vol. 69, no. 11, pp. 7851–7863, Nov. 2021.

[5] X. Liu, B. Lai, B. Lin, and V. C. M. Leung, "Joint communication and trajectory optimization for multi-UAV enabled mobile Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 15354–15366, Sep. 2022.

[6] J. Liu, Y. Shi, Z. M. Fadlullah, and N. Kato, "Space-air-ground integrated network: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2714–2741, 2018.

[7] F. Liu et al., "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022.

[8] Z. Feng, Z. Fang, Z. Wei, X. Chen, Z. Quan, and D. Ji, "Joint radar and communication: A survey," *China Commun.*, vol. 17, no. 1, pp. 1–27, Jan. 2020.

[9] A. R. Chiriyath, B. Paul, G. M. Jacyna, and D. W. Bliss, "Inner bounds on performance of radar and communications co-existence," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 464–474, Jan. 2016.

[10] A. R. Chiriyath, B. Paul, and D. W. Bliss, "Radar-communications convergence: Coexistence, cooperation, and co-design," *IEEE Trans. Cognit. Commun. Netw.*, vol. 3, no. 1, pp. 1–12, Mar. 2017.

[11] K. Meng, Q. Wu, S. Ma, W. Chen, K. Wang, and J. Li, "Throughput maximization for UAV-enabled integrated periodic sensing and communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 1, pp. 671–687, Jan. 2023.

[12] Y. Liu, S. Liu, X. Liu, Z. Liu, and T. S. Durrani, "Sensing fairness based energy efficiency optimization for UAV enabled integrated sensing and communication," *IEEE Wireless Commun. Lett.*, vol. 12, no. 10, pp. 1702–1706, Oct. 2023.

[13] X. Lin et al., "The sky is not the limit: LTE for unmanned aerial vehicles," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 204–210, Apr. 2018.

[14] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Upper Saddle River, NJ, USA: Prentice-Hall, 1996.

[15] M. Razaviyayn, M. Hong, and Z.-Q. Luo, "A unified convergence analysis of block successive minimization methods for nonsmooth optimization," *SIAM J. Optim.*, vol. 23, no. 2, pp. 1126–1153, 2013.